2 THE STONE AGE WILL NOT END FOR LACK OF STONES

In the previous chapter I compared fossil fuels to grandmother's savings hidden under the mattress: a one-time inheritance we've used for an increasingly lavish spending spree for the last two centuries.

The question that haunted my generation, the question planted by that burning-candle poster and watered by *The Limits to Growth*, was simple: when do the savings run out?

I spent years worrying about this, and I wasn't alone. The 1970s oil shocks seemed to confirm our fears. I think that's why *Limits to Growth* sold thirty million copies. A whole generation learned to see the future as a cliff edge, that we were headed toward like mindless lemmings running to their death.

But just like the award-winning Disney documentary showing lemmings running over a cliff turned out to be a cruel fabrication (the filmmakers threw them), (Snopes, 1996) this narrative of humanity heading toward a cliff because of fossil fuels running out was also a fabrication, borne from our fears and the knowledge of journalists that doom scenarios sell. (Robertson, 2023)

Ahmed Yamani turned out to be much closer to the truth. In the year 2000, Sheikh Ahmed Zaki Yamani, Saudi Arabia's oil minister for a quarter century, told an interviewer from The Telegraph: "The stone age did not end for lack of stones, and the oil age will not end for lack of oil." (Brandreth, 2021)

He was echoing a pattern as old as technology itself. The bronze age didn't end because we ran out of copper and tin. The age of whale oil didn't end because we hunted the last whale. The age of horses didn't end because we ran out of oats.

Each era ended because human ingenuity found something cheaper, more convenient, more powerful. And the incumbent industry, confident in its irreplaceability, rarely saw it coming. (Lucas, 2009)

The fossil fuel era is ending the same way. Not with a dramatic depletion, but with a slow economic squeeze. The wells keep producing. For now. The tankers keep sailing. For now. But the financial statements keep deteriorating. We are watching a crash in slow motion.

I will show you the evidence. But first, let me remind you why this matters beyond economics.

2.1 Scarcity Brings Out the Worst in Us

Scarcity activates our tribal instincts. Psychologists have documented this even in infants: when resources are limited, babies as young as thirteen months expect in-group members to favor their own over others. (Bian, 2018) In classic experiments with older children, competition for scarce prizes rapidly generated intense intergroup hostility. (Sherif, 1961) We define our in-group smaller and smaller. We treat out-groups with increasing suspicion, then hostility. We look for scapegoats.

This is not a moral failing. It is an adaptation. For most of human history, scarcity was the normal condition. When the harvest failed, there was not enough for everyone. The question was only who would go without and die. Evolution rewarded those who secured resources for their kin, even at others' expense. (Bourke, 2014)

I think of the Dutch Hunger Winter of 1944-45 that my parents impressed upon me. Twenty thousand Dutch civilians died of starvation while German occupiers confiscated food. The Hunger Winter was brief and local. Impressive for them and me but insignificant on a global scale. But this also happened elsewhere.

The Holodomor was systematic and vast. In 1932-33, Stalin's Soviet regime confiscated grain from Ukraine, sealing borders to prevent escape. Around four million Ukrainians starved to death. (Rudnytskyi, 2015) This was not famine from crop failure. It was famine as policy, engineered by controlling a scarce resource. The grain existed. It was simply taken.

The pattern repeats throughout history. The siege of Leningrad. The Bengal famine of 1943, where multiple factors including wartime policies, Japanese occupation of Burma, and inter-provincial trade barriers led to mass starvation—a distribution problem, as Amartya Sen demonstrated, not an absolute shortage. Mao's Great Leap Forward, which killed tens of millions through centralized control of agricultural production. (Dikötter, 2010) In each case, the horror came not from absolute scarcity but from who controlled the distribution.

Concentrated resources create concentrated power. And concentrated power, history suggests, eventually gets abused.

2.2 Those Who Wield a Scarce Resource Wield Absolute Power

This dynamic translated directly to fossil fuels. When coal powered the Industrial Revolution, the coal barons became kingmakers. When oil replaced coal as the lifeblood of industrial civilization, the pattern intensified beyond anything that came before.

A single commodity, liquid and energy-dense, became the life blood of modern economies. (Oxford Academic, 2018) Whoever controlled the oil controlled the world. The 1973 oil embargo made it visceral. The Arab states turned off the tap, and suddenly the most powerful nations on Earth had to ration gasoline. That's what the control of concentrated energy flows can do.

There are many examples where we literally traded blood for oil. (Sifry, 1991) In 1953, the CIA and British intelligence overthrew Iran's democratically elected government because it had nationalized oil production. They installed the Shah, whose brutality created the conditions for the 1979 revolution, whose aftermath we are still living with. (Britannica, 2024)

The House of Saud made a different bargain. In exchange for American protection, they kept the oil flowing and the prices stable. (Riedel, 2020) They also funded the global spread of Wahhabism, the austere and intolerant strain of Islam that shaped the ideology of al-Qaeda. (European Parliament, 2013) Fifteen of the nineteen September 11 hijackers were Saudi citizens. (9/11 Commission, 2004)

In 2022, Vladimir Putin demonstrated the lesson again. Russian oil and gas had flowed to Europe for decades, a symbol of mutual dependence that was supposed to make war unthinkable. Germany's policy of *Wandel durch Handel*—"change through trade"—was meant to integrate Russia into the European order. (FPRI, 2014) Instead, it made war fundable. The fossil fuel revenues filled Russia's war chest. When Europe objected to the invasion of Ukraine, Putin weaponized winter itself. (NPR, 2022)

Power corrupts, the saying goes, and absolute power corrupts absolutely. Fossil fuel wealth is not quite absolute power, but it is close. The "resource curse" is well documented: countries with oil wealth tend toward autocracy, corruption, and stagnation. (Ross, 2015) The oil does not cause the tyranny directly. But it makes tyranny sustainable. A dictator who controls the oil does not need his people's consent. He does not need their labor. He barely needs them at all.

I expected this power to grow as we headed for the cliff. The logic seemed inescapable: as fossil fuels depleted, whoever controlled the remaining reserves would wield ever greater leverage. The bullies would become more dangerous, not less.

Fortunately, I was wrong.

2.3 Fossil Fuel: From Bully to Bankrupt

I went looking for evidence of the squeeze, and I found it. But the squeeze was working in the opposite direction from what I expected. The fossil fuel titans weren't getting richer. They were bleeding money. Not all of them, not every quarter, but the trend was unmistakable once you knew where to look. The industry that had controlled the world for a century was quietly becoming insolvent.

Not because the oil ran out. The proved reserves on the balance sheets looked as robust as ever. The tankers kept sailing. The pipelines kept flowing. Something changed in the economics. Something inevitable that is slowly driving the former bullies and kings of the earth toward bankruptcy.

2.4 Coal: Cathedrals Built to Be Abandoned

First I turned to coal. We theoretically have centuries' worth in the ground. It's technically the simplest to extract: no drilling, no fracking, no deepwater platforms. Just dig it up. Coal has generated more electricity than any other source for over fifty years. (IEA, 2025)

And coal is being massacred.

2.4.1 The Dutch Mistake

In 2015, three major European utilities, RWE, Uniper, and Engie, commissioned brand-new coal plants in the Netherlands. These were not aging relics limping toward retirement. They were the newest, most efficient coal plants in Europe, representing over €6 billion in investment. State-of-the-art technology. Decades of expected operation.

Within one year, the collective book value had fallen to €2.5 billion. RWE's Eemshaven plant alone cost €3 billion to build; RWE later wrote down its entire Dutch conventional fleet to €1.3 billion. Analysts at the Institute for Energy Economics and Financial Analysis estimated the plants might actually be worth closer to €400 million each. (IEEFA, 2016)

The utilities are now suing the Dutch government for compensation after a 2030 coal ban. RWE claims €1.4 billion; Uniper claims nearly €1 billion. The courts have been unsympathetic. They ruled that the government had a legitimate interest in meeting climate commitments.

Economics, even more than climate policy, is spelling doom for coal. The newest, most efficient coal plants in Europe were economically obsolete before the ribbon-cutting ceremony was over. The companies that built them, with all their expertise and due diligence, made multi-billion euro mistakes.

2.4.2 The Birthplace of Coal Shuts It Down Completely

On September 30, 2024, at midnight, the Ratcliffe-on-Soar power station in Nottinghamshire finished its final shift. (PBS, 2024) With that, 142 years of coal-powered electricity in Britain came to an end.

Britain invented coal-fired power generation. The first public power station, at Holborn Viaduct in London, opened in 1882. At its peak in 1920, more than one in twenty British workers labored in coal mines. (Ritchie, 2019) For more than a century, coal was synonymous with British industry, British power, British identity. As recently as 2012, coal generated 40% of British electricity.

Twelve years later: zero.

The plant manager, Peter O'Grady, told reporters: "When I started my career 36 years ago, none of us imagined a future without coal generation in our lifetimes." (CBS News, 2024)

Britain became the first G7 nation to phase out coal power. UK coal demand in 2024 fell to its lowest level since 1666—the year of the Great Fire of London. (Carbon Brief, 2025) The country that started the Industrial Revolution walked away from the fuel that made it possible.

Coal was not replaced by gas. It was replaced by wind, which grew 315% over the period, along with solar and simple efficiency gains. (Ember, 2024) Power sector emissions fell 74% from their 2012 levels. (Ember, 2024)

2.4.3 Germany Pays Plants to Close

In December 2020, Germany held its first auction for coal plant closures. The government had set aside €165,000 per megawatt as compensation for operators who agreed to shut down early.

The auction was massively oversubscribed. Operators accepted an average of just €66,000 per megawatt, less than half the ceiling. Some bids came in as low as €6,047 per megawatt. (Clean Energy Wire, 2020)

Read that again. The government offered to pay companies to close their coal plants. The companies competed to close first. They fought for the privilege of exiting coal.

The fear that phase-out would be expensive proved backwards. Keeping coal running was the expensive option. The companies were eager to stop the bleeding.

Uniper's Datteln 4 plant opened in 2020 as "one of the most modern power plants of its kind in the world." (Uniper, 2020) It was expected to operate until Germany's 2038 coal phase-out deadline. Five years later, Uniper sold it. (Rigzone, 2025) The company now aims to exit coal entirely by 2029. (Uniper, 2023)

2.4.4 China: The Counterintuitive Case

"But what about China?" people ask. "They're still building coal plants."

They are. And that makes what's happening even more striking.

In 2000, coal generated about 80% of China's electricity. (IEA, 2024) By mid-2025, that share had fallen to 51%, a historic low. (CREA, 2025) The decline accelerated in 2024, when clean energy met 84% of new electricity demand. (Ember, 2025) In the first half of 2025, for the first time, growth in renewables didn't just supplement coal but actually drove a 2% decline in fossil generation. (Ember, 2025)

The numbers are staggering. In 2024, China installed 277 gigawatts of solar, more than twice the total solar capacity the United States has ever built. (CREA, 2025) China's 2030 target of 1,200 gigawatts of combined wind and solar was reached in 2024, six years early. (Cipher News, 2024)

Why does this matter? Because clean energy has become too economically important to abandon. It now contributes over 10% of China's GDP and accounts for more than a quarter of all economic growth. (Carbon Brief/CREA, 2025) The "new three" industries, electric vehicles, batteries, and solar, are what keeps China's economy expanding. Without clean energy growth, China's 2024 GDP growth would not have been 5%, right on target, but just 3.5%. (Carbon Brief/CREA, 2025)

Yes, China continues to build coal plants, the legacy of a 2022-2023 permitting surge driven by energy security fears. But permits have since fallen 83%. (CREA/GEM, 2024) Many of these new plants will be like the Dutch coal cathedrals: built but barely used. Coal plant utilization is already falling. The question is not how many coal plants China is building. The question is how much coal they will actually burn.

The country building the most coal is also the country where coal's share is declining fastest. The economics are winning, also in China.

2.4.5 India: The Arithmetic Becomes Undeniable

India's Parliamentary Standing Committee on Energy identified 40 gigawatts of stranded coal projects. (IEEFA, 2018) Solar tariffs in India have fallen below the fuel cost alone of running existing coal plants. Not the total cost, the fuel cost. It is now cheaper to build new solar than to buy coal for the plants that are already built. (Shrimali, 2020)

NTPC, India's state-owned power giant and largest thermal generator, announced it would not pursue any new greenfield coal projects. The company is pivoting to 60 gigawatts of renewable capacity by 2032. (Business Standard, 2021) No new coal-fired power plants have been announced in India in the past twelve months.

A 2025 study in Nature Communications calculated that aggressive coal retirement in India could yield net economic gains of $171 billion in Chhattisgarh and $110 billion in Uttar Pradesh alone. (Garg, 2025) Early closure would reduce air pollution deaths and save costs that exceed the value of the stranded assets by a factor of four. (Nearly 1.7 million people die each year in India from air pollution, with fossil fuels responsible for more than 750,000 of these deaths. (Lancet Countdown, 2025))

India doesn't have to choose between development and decarbonization. Once you understand what is happening, you will see that maximum progress on both fronts is happily aligned.

2.4.6 The Global Pattern

These are not isolated anecdotes. Ember research shows coal power has halved across OECD countries since peaking in 2007. (Ember, 2024) According to Carbon Tracker, nearly 80% of coal plants in the European Union were losing money in 2019, before COVID, before the 2022 energy crisis. (Carbon Tracker, 2019)

The pattern holds from Newcastle to New Delhi: coal isn't dying because we're running out. It's dying because something better came along.

2.5 Gas: The Transition Fuel That Wasn't

Natural gas was once hailed as the transition fuel. It emits about half as much carbon dioxide as coal when burned for electricity. Cleaner burning. Fewer particulates. The 'sensible compromise' between dirty coal and uncertain renewables. Then came fracking.

2.5.1 The Google of Oil and Gas

Chesapeake Energy pioneered hydraulic fracturing in the American shale formations. The company was once valued at $37 billion. (CBS News, 2020) The press called it "the Google of oil and gas." Its CEO, Aubrey McClendon, graced the covers of business magazines. Chesapeake almost single-handedly made the United States the world's largest natural gas producer.

In June 2020, Chesapeake filed for bankruptcy with $9 billion in debt. (CNN, 2020)

The details are instructive. Between 2010 and 2012 alone, Chesapeake spent $30 billion more in drilling and leasing than it earned from operations. (NBC News, 2020) The company produced gas at a phenomenal rate, but the production never covered the costs.

When auditors examined the books, they found curiosities. Some $110 million had been spent on two corporate parking garages. There was a wine collection hidden behind a broom closet. (NBC News, 2020)

Chesapeake was an extreme case, but it was not an outlier. The entire U.S. shale gas industry followed a similar pattern: record production, capital destruction. The gas flowed. The money vanished.

Fracking also had another problem. Methane, the main component of natural gas, is a potent greenhouse gas, about 82 times more warming than CO_2 over twenty years and about 30 times more over a century. (IPCC, 2021) Studies found that methane leaking during drilling and transport could make fracked gas as damaging to the climate as coal. (Howarth, 2011) So much for the transition fuel.

2.5.2 Gazprom's Artificial Resurrection

In Russia, the state-controlled gas giant Gazprom reported a remarkable turnaround in 2024: a profit of 1.2 trillion rubles, up from a loss of 629 billion rubles the year before. (Moscow Times, 2025)

The numbers deserve scrutiny.

The 2024 "profit" included 500 billion rubles from Gazprom Neft, the company's oil subsidiary. The gas business itself made only 719 billion rubles. Meanwhile, the 2023 loss had included a 1.3 trillion ruble windfall tax to fund military spending and 1.15 trillion rubles in write-offs of European assets seized after sanctions. (Vakulenko, 2025)

Strip away the one-time charges and the oil subsidiary's contribution, and what remains is a gas business barely breaking even.

More telling: Gazprom's net debt service costs quintupled between 2019 and 2024, from 171 billion rubles to 719 billion. (Vakulenko, 2025) The company is using oil profits to subsidize gas operations while debt climbs toward dangerous levels. This is not recovery. It is financial triage.

2.6 Oil: The Last Man Standing Isn't Standing

I expected oil to be the last holdout. It is the most valuable fossil fuel, the most versatile, the hardest to replace. Liquid gold. The stuff that keeps armies moving. If any fossil fuel would survive the transition, surely it would be oil.

2.6.1 The $300 Billion Bonfire

From 2010 to 2019, the U.S. shale oil industry generated negative $300 billion in cumulative free cash flow. (Deloitte, 2022) Over 190 companies filed for bankruptcy. (Deloitte, 2020) More than $450 billion in invested capital was written off. (Deloitte, 2020)

This happened while oil averaged over $80 per barrel for much of the period. At that oil price they should have been printing money. Instead they burned it. According to Deloitte, 30% of shale operators were technically insolvent at $35 per barrel. (CNN/Deloitte, 2020)

The industry produced record volumes of oil but also destroyed record volumes of money.

2.6.2 The Supermajors' Quiet Hemorrhage

Shale drillers, you might argue, were always gamblers. Small operators, undercapitalized, chasing the next boom. Surely the supermajors would show how it's done. ExxonMobil. Shell. Chevron. BP. TotalEnergies. A century of experience. The deepest pockets in the business.

The Institute for Energy Economics and Financial Analysis examined their SEC filings and found a different story. (IEEFA, 2020)

From 2010 through 2019, the five supermajors collectively generated $340 billion in free cash flow. That sounds robust until you learn that over the same period, they paid $556 billion to shareholders in dividends and buybacks.

The gap: $216 billion. More than $200 billion that they paid out but didn't earn. (IEEFA, 2020)

Extend the analysis to 2005 through 2020, and the cumulative deficit reaches $325 billion. The supermajors funded this shortfall through a combination of new debt, adding over $210 billion in long-term borrowing, and asset sales. (IEEFA, 2023) BP and Shell each sold roughly $70 billion in assets over the decade. (IEEFA, 2020)

In only three of those fifteen years did the supermajors generate a cash surplus.

2.6.3 Prices Up, Profits Down

The IEEFA documented a striking comparison between the second quarter of 2023 and the third quarter of 2021—two periods with comparable oil prices that allowed them to isolate the effect of rising costs. (IEEFA, 2023) Oil prices had gone up but the supermajors' combined free cash flow was down more than 20%. The industry made less money while selling its product for more.

This is what rising extraction costs look like when translated into dollars. When costs rise faster than revenues, margins compress even as prices increase. The headline numbers look stable. The underlying economics are deteriorating.

2.6.4 The Confessions

In mid-2020, as the pandemic cratered demand, the supermajors finally started admitting what their balance sheets had been showing for years.

BP wrote down $17.5 billion in assets. Shell followed with up to $22 billion. (Inside Climate News, 2020) Both companies simultaneously lowered their long-term oil price assumptions to $55-60 per barrel, an acknowledgment that the future they had planned for was not coming. (World Oil, 2020)

Wood Mackenzie, the energy consultancy, called it "a message about stranded assets." (Wood Mackenzie, 2020) The world's largest, most experienced oil companies were admitting their own forecasts had been wrong.

Russia-related write-offs alone cost foreign oil and gas companies nearly $58 billion in 2022, with BP taking the largest hit at $25.5 billion. (Interfax, 2023)

2.6.5 Europe's Refinery Extinction

The decline is visible in industrial infrastructure. Europe lost 400,000 barrels per day of refining capacity since early 2024: five refineries closed in two years, including three since March. (Argus, 2025) High taxes, climate rules, and smaller-scale sites are making European refineries less competitive than those in other regions.

Long-term naphtha and diesel demand forecasts have collapsed under the weight of electric vehicle adoption. (Hydrocarbon Engineering, 2025)

These aren't marginal adjustments. They're structural amputations. European refineries must now reinvent themselves or close.

2.7 Sidebar: Why Costs Keep Rising (EROI)

The financial deterioration traces back to physics. Energy Return on Investment, or EROI, is the ratio of energy obtained to energy expended. Think of it as the profit margin of physics.

2.7.1 The Concept

In the early days of oil, EROI was spectacular. The oil at Spindletop in 1901 essentially pushed itself out of the ground. Drill a hole, stand back, let gravity and pressure do the work. For every barrel of oil equivalent invested in extraction, you might get back a hundred barrels. An EROI of 100:1.

Those days are gone. Conventional oil fields still producing today have EROIs around 18:1 to 20:1. Respectable, but far from the early gushers. New production increasingly comes from unconventional sources: deepwater drilling (EROI around 10:1), Canadian oil sands (around 5:1), tight oil from shale (5:1 to 10:1). (Hall, 2014)

The easy oil came first. This is not conspiracy or mismanagement. It is common sense. You don't start with the hardest rock when easier rock is available. But there is nothing we have looked for more tenaciously on earth than oil. So we already found most of the easy rock. What remains is deeper, dirtier, more energy-intensive to extract.

2.7.2 The Alberta Tar Sands Example

Take the Alberta oil sands. Industry sources often claim an EROI of about 5:1, meaning you get five barrels of energy for every barrel you invest. That sounds acceptable. But this figure is measured "at the mine mouth," and it hides enormous losses downstream. (Hall, 2014)

The extraction process uses vast quantities of natural gas for heating and for generating electricity to run pumps. Producing that natural gas has its own energy costs. Converting gas to electricity loses about half the energy. Transporting the heavy bitumen requires diluting it with lighter hydrocarbons (which themselves required energy to produce) or upgrading it on-site (more energy). Refining the resulting crude into gasoline burns more energy still. Trucking the gasoline to your local station: more losses.

By the time tar sands oil reaches your fuel tank, a reasonable accounting suggests the net energy gain may largely evaporate. As Charles Hall of SUNY has noted, the EROI for oil sands would fall closer to 1:1 if the full life cycle, including transportation, refinement, and end-use efficiency, is accounted for. (InsideClimate, 2013)

2.7.3 The 4.6:1 Threshold

A 2024 study in Nature Energy examined this problem systematically. The researchers asked: what EROI at the extraction point is actually needed for fossil fuels to deliver net energy to society, accounting for all the losses in processing, refining, and distribution? (Aramendia, 2024)

Their answer: 4.6:1 at the mine mouth, at minimum.

Below this threshold, the energy source may be profitable in dollar terms (if energy inputs are cheap and outputs are expensive) while actually draining energy from the economy. Money can flow in circles even when net energy doesn't.

This explains an odd phenomenon: tar sands projects that receive billions in subsidies despite questionable energy returns. The gas used for extraction is cheap because it's abundant and undertaxed. The gasoline produced sells for more because it's in demand. On a spreadsheet, this looks profitable. In physical terms, society is putting in more energy than it gets back.

2.7.4 Why Renewables Are Different

For fossil fuels, the EROI at the mine mouth and the EROI at the point of use are drastically different, because combustion-based energy systems have large losses at every step.

For renewables, this gap largely disappears. A solar panel factory powered by solar panels doesn't lose 50% of its energy in conversion. An electric car charged by solar doesn't waste three-quarters of its energy as heat the way a gasoline engine does. The whole chain becomes more efficient.

This means a solar installation with an EROI of even 10:1 at the panel delivers far more useful energy to society than a fossil fuel source with the same nominal EROI. And current estimates put solar's EROI at 12:1 and rising, wind at 24:1 and rising. (Aramendia, 2024) As manufacturing scales and efficiency improves, these numbers continue to climb.

Practically speaking EROI has become a non issue for renewables but is increasingly becoming a show stopper for fossil fuels, especially unconventional fossil fuels.

2.7.5 The Dallas Fed Evidence

The falling EROI shows up directly in producer surveys. The Dallas Federal Reserve asks oil producers each quarter what price they need to profitably drill a new well. (Dallas Fed, 2025)

In 2019, the average breakeven price in the United States was $50 per barrel. By early 2025, it had risen to $65 per barrel. That's a 30% increase in six years, faster than general inflation. (Dallas Fed, 2025)

One survey respondent put it plainly: "The U.S. oil cost curve is in a different place than it was five years ago. $70 per barrel is the new $50." (Dallas Fed, 2025)

Enverus Intelligence Research projects breakeven prices could hit $95 per barrel by the mid-2030s as producers work through their remaining prime inventory. (Enverus, 2025)

2.8 Sidebar: "Profitable" Isn't What You Think It Is

Understanding the slow squeeze requires distinguishing between two concepts that sound similar but measure different things.

Reported profit (or net income) equals revenue minus expenses as defined by accounting rules. These expenses include operating costs, depreciation, interest, and taxes. By this measure, the supermajors report profits most years.

Free cash flow equals actual cash generated from operations minus capital expenditures. It's the money left over after paying for everything needed to maintain the business, including drilling new wells to replace depleting ones.

This distinction matters because oil companies must continuously invest just to stay in place. A typical shale well loses 50-75% of its production in the first year alone. (Gong, 2016) Conventional fields decline more slowly but still decline. Without constant drilling, income falls.

This capital spending is enormous. For the supermajors, it typically runs $15-25 billion per year, each. Accounting rules spread this cost over many years through depreciation, which smooths reported profits. But the cash goes out the door immediately.

A company can report accounting profits while spending more cash than it generates. The income statement shows health. The cash flow statement shows hemorrhage. The balance sheet, over time, shows the truth: debt rising, assets falling, the company slowly consuming itself.

This is not theory but it showing up in reality. From 2010 to 2019, the five supermajors:

- Reported cumulative net income of roughly $400 billion
- Generated free cash flow of $340 billion
- Paid shareholders $556 billion

(IEEFA, 2020)

The gap between cash generated and cash distributed was filled by borrowing and asset sales. This is sustainable for a while. It is not sustainable forever.

When an industry consistently pays out more than it earns, and funds the difference with debt and divestitures, it is not investing in its future. It is liquidating while keeping up appearances.

2.9 The Pyramid Scheme Dynamics

Across all three fossil fuels, the same pattern emerges. Costs rise faster than revenue. Dividend checks keep coming, but debt grows and assets shrink. The headline numbers look stable while the foundation rots. And it's not even the fault of the CEOs, because this is what they are paid to do.

CEOs are paid to deliver shareholder value. Shareholder value equals stock price plus dividends. Cutting dividends crashes stock prices. So their salary depends on keeping dividends going, even when cash flow doesn't support them. This will go on until the scheme collapses. Any CEO who doesn't want to play along will be replaced, and by the time the consequences arrive they will be retired anyway. What would you do?

I am not accusing any individual of fraud. As often it's the structure itself that produces the behavior. Every CEO's incentive is to maintain confidence until after they've collected their bonus. The industry runs a slow-motion liquidation disguised as stability.

But once the confidence falters, the downward spiral can lead to a crash. Oil fields that turn out to be economically unrecoverable stranded assets get written off. Credit ratings slip, so borrowing costs rise. Debt service becomes unsustainable, so dividends must be cut. Dividend cuts crash the stock. Stock crashes get executives fired. New executives inherit a gutted company.

This is not speculation about the future. It is simple extrapolation of what we are seeing in the financial report, and explained by the physics of rising EROI combined with cheaper alternatives for fossil fuels.

In 2020, Wood Mackenzie estimated that price crashes and demand destruction removed $1.6 trillion in value from the world's upstream oil and gas industry. (Wood Mackenzie, 2020) This was not called a crash. It was called "fundamental change hitting the entire oil and gas sector." The language of financial analysts pussyfoots around the problem because it's a problem too large to face head on.

2.10 The Stone Age Didn't End For Lack of Stones

I feel no sorrow for Putin as the funding for his war evaporates. For the House of Saud as the basis for their bargain with religious extremists disappears. For the oil majors that spread doubt about climate change for decades while their own scientists knew the truth. Humanity has profited immensely from fossil fuels these last two centuries, but we can say goodbye to the politics and scarcity surrounding them without remorse.

Sheikh Yamani's warning came true, though probably not the way he expected. The oil age is ending not because wells are running dry but because the economics are running dry. The reserves on the balance sheets are still there. Economically speaking, most of it is becoming stranded assets. Because something better than stones came along.

I went looking for doom and found, to my surprise, a kind of liberation. I started with the thought of a scarce resource creating a cliff we were running toward. A centrally controlled resource that created tyrants in the process. For decades I feared that fossil fuel depletion would trigger resource wars, authoritarian consolidation, civilizational collapse.

Instead, I find the power of and need for fossil fuels are waning. I underestimated human ingenuity. It almost seems like we are inventing our way out of the trap we invented our way into.

But if fossil fuels are dying, what replaces them?

For a long time, I thought I knew the answer. A technology that seemed to offer unlimited clean power. Energy too cheap to meter, drawn from the heart of the atom itself.

I wanted nuclear power to be the solution. The physics seemed so elegant, the energy density so remarkable. Then I looked at the actual numbers.